ovid GT Reference Guide

Ground temperature loggers: what the app does to every reading, why, and how to check it. For GT 1.7.0 (ovid Core 1.14.0). Settings are quoted with their defaults; change them in the app (Settings, Quality rules) or in `CFG` at the top of `gt_pipeline.R`.

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Quick start

1. Put the logger files in the data folder, in any layout (for example `data/rbr/2024`, `data/cryogeek/2024`, `data/hobo/…`, `data/em50/…`, `data/campbell/…`). Keep each logger's own file *and* its exports: an RBR `.hex` with its Ruskin `.txt`, a GeoPrecision `.f2b` with its FG2-Shell `.csv`, a `.hobo` with its HOBOware `.csv`.
2. Put the registry workbook (a sheet with Name, Type and SN columns), any borehole metadata workbooks and field notes (`.docx`) in the metadata folder.
3. Start the app (`Open GT (Mac).command`, or `Ovid_GT.bat` on Windows) and set the data, metadata and project folders on **Home › Data source** (each has a **Browse…** button).
4. Run the **Set-up check**. Fix anything in red first.
5. Press **Run pipeline**. The first run seeds `sites.csv` and `sensors.csv` from the registry.
6. Open **Sensors and depths**: every sensor needs a site and a depth. Depths marked *default* or *template* are assumptions; replace them with measured depths.
7. Read **Raw files** for bundles that *need export* (a `.f2b` or `.hobo` with no readable export) and **Clock checks** for resets and drift. The `.f2b` and `.hobo` files themselves cannot be read: export them with FG2_Shell (GeoPrecision) and HOBOware Pro's **Bulk Export** (Onset), which write a CSV beside each file, then run again.
8. Run again. Read **Sensor health**, the flags on **Plot and explore** (QC view) and the between-site check on **Compare sites**.
9. Export data and reports (HTML, PDF or both) from **Reports and export**. Every run also writes `output/report.html`.

Only loggers whose files or set-up changed are processed again (setting `incremental = TRUE`, cache in `<project>/cache`). **Rescan** on the Raw files page reads the selected files again and rebuilds only their loggers and sites; everything else is reused. Adding one new logger does not make the other loggers run again.

What happens to the data, in order

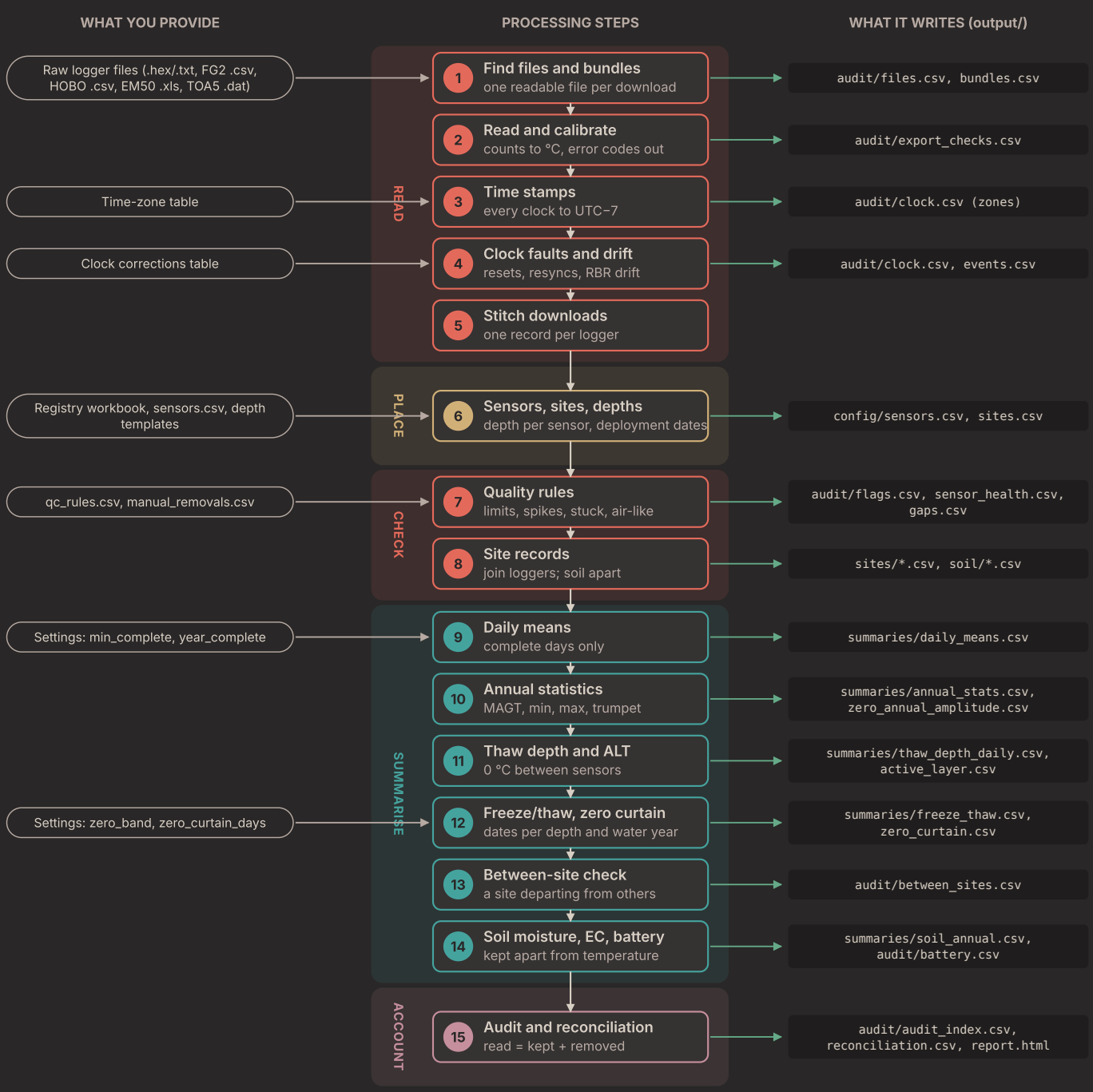
#	Step	What changes	Logged in
1	Find files and bundles	Each download becomes a bundle; the best readable file is used; byte-identical copies skipped	`audit/files.csv`, `bundles.csv`
2	Read and calibrate	RBR `.hex` counts converted to °C; values the logger marks as errors removed	`audit/export_checks.csv`, audit trail (`sensor_error`)
3	Time stamps	Every clock moved to UTC−7	`audit/files.csv` (zone, zone_why)
4	Clock faults and drift	Impossible times, clock resets and superseded readings removed; hand shifts applied; RBR drift optional	`audit/clock.csv`, `events.csv`
5	Stitch	One record per logger; repeats kept once; the newest download wins a disagreement	audit trail (`duplicate_value`, `overlap_conflict`)
6	Sensors and depths	Each value gets a site and a depth; values outside the deployment dates removed	`config/sensors.csv`, audit trail (`outside_deployment`)

7	Quality rules	Limits and spikes removed; stuck, air-like, pre-install and manual periods flagged or removed	`audit/flags.csv`, `sensor_health.csv`, `gaps.csv`
8	Site records	Loggers that replaced each other joined (later deployment wins); soil apart	`sites/`, `soil/`, audit trail (`site_overlap`)
9	Daily means	Mean, min, max per site, depth and day; incomplete days marked	`summaries/daily_means.csv`
10	Annual statistics	MAGT, min, max, range per depth and year	`annual_stats.csv`, `zero_annual_amplitude.csv`
11	Thaw depth and ALT	Depth of 0 °C each day; deepest thaw each year where permafrost is present	`thaw_depth_daily.csv`, `active_layer.csv`
12	Freeze/thaw	Freeze and thaw dates per depth and water year; zero-curtain periods	`freeze_thaw.csv`, `zero_curtain.csv`
13	Between sites	Flags days on which a site departs from the others at the same depth (never removes)	`audit/between_sites.csv`, `flags.csv`
14	Soil and battery	Moisture and conductivity summarised apart; battery to its own table	`soil_annual.csv`, `audit/battery.csv`
15	Audit	Every removal indexed; counts reconciled per logger channel	`audit/audit_index.csv`, `reconciliation.csv`

Every value read is accounted for: **values read = values in the output + values removed**, per logger and channel. The run stops if that does not hold. Raw files are never changed.

The pipeline at a glance

Five stages, fifteen steps. Inputs on the left enter at the step that uses them; files on the right are written at the step that makes them. Each step passes the full record down; nothing leaves without an entry in the audit trail.



One reading through the pipeline

A real reading from a bog borehole with a GeoPrecision FG2 logger and a four-node thermistor chain (nodes at 0.05, 0.55, 0.85 and 1.00 m; 30-minute logging). The reading is node 2, at 0.55 m, at noon on 15 August 2022. Site and logger identifiers are left out.

In the raw file

```
NO,TIME,#1((unk.)),#2((unk.)),#3((unk.)),#4((unk.)),HK-BAT(V)
53576,15.08.2022 12:00:00,20.056,15.963,12.342,9.161
```

Step	What happens to this reading	Value after the step
1 Files	The `.f2b` cannot be read; its FG2-Shell `.csv` in the same bundle is used (MD5 recorded). The `.log` is a companion.	row 53576, column `#2` → channel `T2`
2 Read	FG2-Shell already wrote °C; no `(NoReply)` on this row	value = 15.963 °C

3 Time	No zone in the file and no time-zone table row: the logger clock is taken as UTC−7	2022-08-15 12:00:00, shift 0 h
4 Clock	After 2000, not in the future, not written after a clock reset or resync (this logger had 10 superseded readings per channel elsewhere)	kept
5 Stitch	Only one download holds this time	unchanged
6 Depth	Sensor `T2` of this serial has depth 0.55 m from the registry, deployment from 2019-08-07	site = the borehole, depth_m = 0.55, depth_source = registry
7 QC	Inside −50 to 50 °C. Spike threshold = max(1, 10 × 1.4826 × MAD of steps) = max(1, 0.51) = 1 °C; the step in is +0.130 and out −0.153: no spike. Not flat; day range 2.8 °C (air-like needs over 15 °C)	kept, no flag
8 Site	Only one logger at this site and depth	column `T_0.55m`
9 Daily mean	One of 48 readings that day (needs ≥ 0.8 × 48 = 38.4)	mean 16.750, min 15.298, max 18.098, complete
10 Annual	One of 365 complete days of calendar year 2022 at 0.55 m	mean 3.139, min −10.471 (6 Dec), max 16.868 (14 Aug), range 27.338
11 Thaw depth	All four depths above 0 °C that day	thaw_m blank: thawed below the deepest sensor
12 Freeze/thaw	Water year 2023 starts on 1 Oct 2022, after this day; the reading counts in water year 2022	see the table below for water year 2023
13 Between sites	Five sites have a node at 0.55 m. Others' median 15.910; offset +0.840; usual offset (31-day running median) +0.622; departure +0.219 °C, far below max(6 × 0.42, 1.5) °C	not flagged
15 Audit	Channel T2: 90,445 values read = 89,724 in the output + 721 removed (708 outside deployment, 10 clock resync, 3 spikes)	reconciles

What the outputs look like

sites/<site>.csv (wide: one column and one flag column per depth):

timestamp_utc-07	T_0.05m	T_0.05m_flag	T_0.55m	T_0.55m_flag	T_0.85m	T_0.85m_flag	T_1m	T_1m_flag
2022-08-15 11:30:00	19.532		15.833		12.338		9.160	
2022-08-15 12:00:00	20.056		**15.963**		12.342		9.161	
2022-08-15 12:30:00	20.535		16.116		12.350		9.161	

summaries/daily_means.csv:

site	depth_m	date	temp	tmin	tmax	n	expected	complete
…	0.05	2022-08-15	18.991	14.609	26.165	48	48	TRUE
…	0.55	2022-08-15	16.750	15.298	18.098	48	48	TRUE
…	0.85	2022-08-15	12.340	12.248	12.426	48	48	TRUE
…	1	2022-08-15	9.161	9.147	9.176	48	48	TRUE

summaries/annual_stats.csv (calendar 2022):

depth_m	year	mean	min	max	range	days	date_min	date_max	complete
0.05	2022	3.783	−10.577	20.036	30.613	365	2022-12-06	2022-08-14	TRUE
0.55	2022	3.139	−10.471	16.868	27.338	365	2022-12-06	2022-08-14	TRUE
0.85	2022	3.015	−5.905	12.741	18.646	365	2022-12-06	2022-08-27	TRUE
1	2022	3.829	−0.272	9.871	10.143	365	2022-12-06	2022-09-06	TRUE

summaries/freeze_thaw.csv (water year 2023, October 2022 to September 2023):

depth_m	first ≤ 0	last ≤ 0	first > 0 after	days frozen	days thawed	days zero curtain	mean
0.05	2022-10-23	2023-04-26	2023-04-27	168	192	5	4.725
0.55	2022-11-01	2023-04-26	2023-04-27	163	193	9	3.676
0.85	2022-11-05	2023-05-23	2023-05-24	146	156	63	3.305
1	2022-12-06	2022-12-06	2022-12-07	1	364	0	4.288

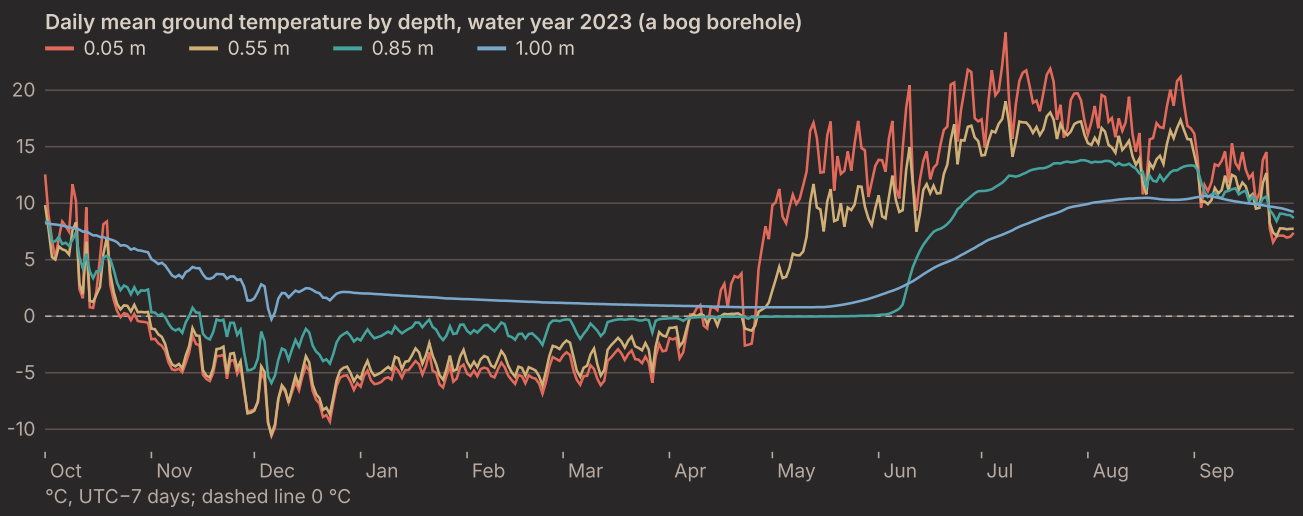
summaries/active_layer.csv (2022): `alt_m` blank, result *no permafrost within the sensors: thawed below the deepest sensor*. The 1.00 m node went below 0 °C on one day only (−0.27 °C), so no depth stays frozen all year; this borehole has seasonal frost, not permafrost, within its 1 m.

audit/flags.csv (the same logger, at the start of the record):

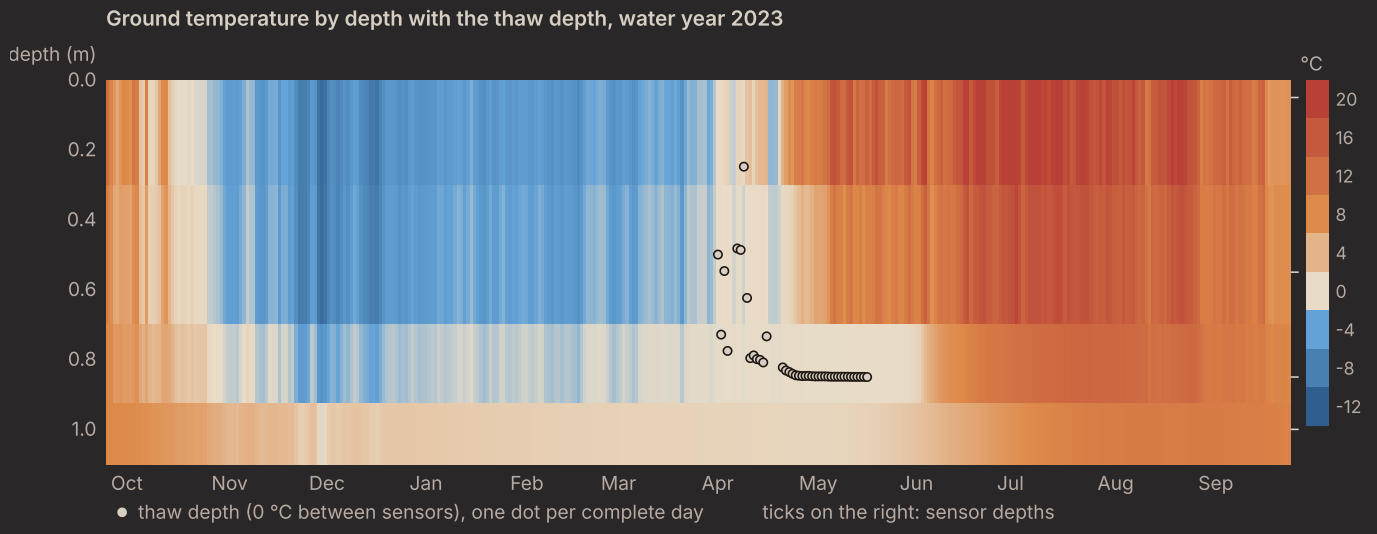
channel	start	end	records	check	detail
T2	2019-08-07 00:00:00	2019-08-08 20:00:00	45	pre_install	at the start of the record every sensor reads the same temperature: the string was probably not in the ground yet

What the plots look like

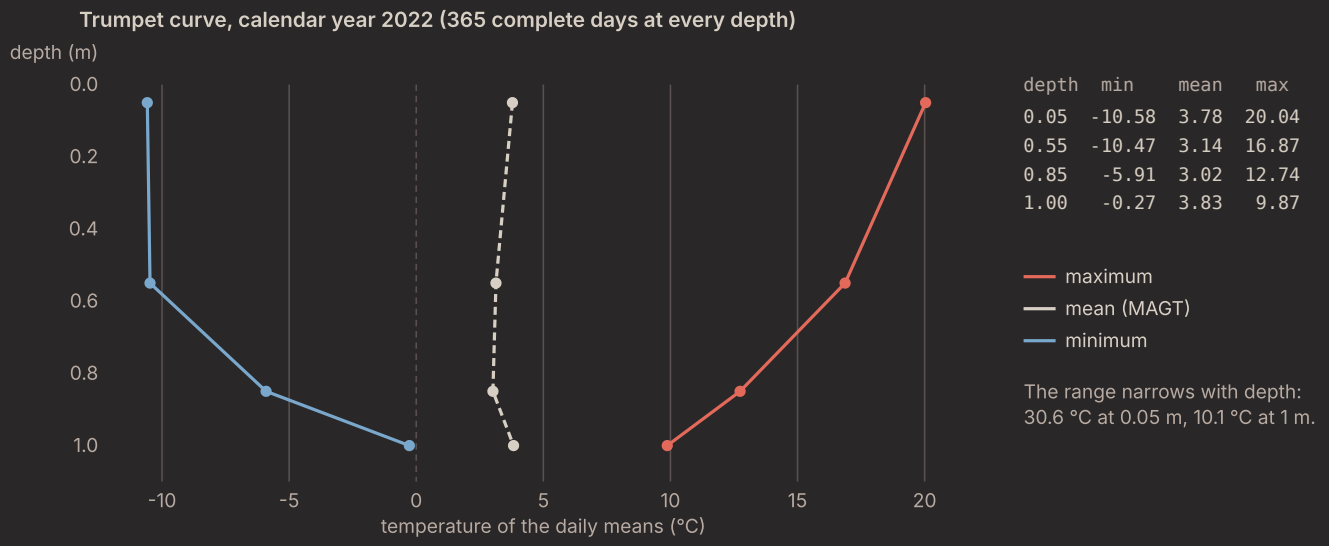
The same borehole drawn as the app shows it on Plot and explore and in the site report.



The surface node (red) follows the air with a lag; the 1.00 m node (blue) barely leaves 0–5 °C. From late April to early June the 0.85 m node stays a few hundredths of a degree below zero (a 40-day zero curtain) while the 0.55 m node above it has already warmed.



One band per sensor, each reaching halfway to its neighbours. The light dots are the daily thaw depth: they appear in spring when the top is thawed and a deeper node is still at or below 0 °C, and settle just above 0.85 m (0.848 m on 10 May). They are absent in summer (thawed below 1 m) and in winter (frozen at the surface).



depth	min	mean	max
0.05	-10.58	3.78	20.04
0.55	-10.47	3.14	16.87
0.85	-5.91	3.02	12.74
1.00	-0.27	3.83	9.87

The trumpet curve plots each depth's annual minimum, mean and maximum of the daily means. The envelope narrows with depth; where it would close to under 0.1 °C is the depth of zero annual amplitude (not reached here).

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1. Finding files and bundles

Every file under the data folders is identified by its content (not its name), with size, MD5 checksum, serial and model. Byte-identical copies (same MD5) are listed as `duplicate_file` and skipped. Files of one download (the logger's own file plus its exports, matched by type, folder, serial and file name) form a **bundle**; the best readable member is read: RBR `.hex` first, then the Ruskin `.txt` or `.xls`.

Logger	Read from	Not readable (kept as identity only)
RBR XR-420 / concerto	`.hex` (calibrated from its header); else Ruskin `.txt` (UTC) or `.xls`	`.rsk`
GeoPrecision FG2	FG2-Shell `.csv`; else the `.f2b` itself, decoded by GT	`.log` (companion)

Onset HOBO U12	HOBOware `.csv`	`.hobo` (header gives serial, model and zone)
METER EM50	ECH2O Utility `.xls`	—
Campbell Scientific	TOA5 `.dat`	mixed-array (CR10X) and binary tables

Bundle status	Meaning
ok	A readable file is used
needs_export	Only an unreadable file (for example an `.f2b` with an empty CSV). The advice says what to export, e.g. *Open the .f2b in FG2-Shell and export it as .csv into the same folder*
unsupported	A Campbell mixed-array or binary table: convert to TOA5 with CardConvert
water_logger	A HOBO water-level (U20) or conductivity (U24) logger: a pressure or conductivity channel in its CSV, or that model in its `.hobo`. Read by WLR, not GT; nothing to export
raw_not_used	A readable logger file (an `.f2b`) whose FG2-Shell export from the same download is read instead

An `.f2b` with no FG2-Shell export is decoded by GT (`R/f2b.R`), matching FG2-Shell's export row for row: channels `#1, #2 …` become `T1, T2 …`, (NoReply) is a logger error, the battery is its own variable and a new parameter set is an event. Times are the logger clock as FG2-Shell shows them; a clock that was never set reads from 1998-01-01 and those records are removed as impossible times. An `.f2b` that does not start with an FG2 marker (for example a file saved under the wrong name) is an unrecognised format; one that starts like an `.f2b` but cannot be decoded still needs an export.

Override decisions with File rules (pattern, action, serial, reason, author), e.g. ignore a test file or set a serial. Outputs: `audit/files.csv`, `audit/bundles.csv`.

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2. Reading and calibration

RBR .hex: counts to temperature

The `.hex` holds raw 24-bit counts per channel and the four calibration coefficients of each thermistor in its header. The app applies RBR's Steinhart–Hart style polynomial:

```
r = count / 2²⁴
L = ln( (1 − r) / r )
T (°C) = 1 / (c0 + c1·L + c2·L² + c3·L³) − 273.15
```

Example. Coefficients from an XR-420 header: c0 = 0.00432304, c1 = −0.000277840, c2 = 0.00000362266, c3 = −0.000000125537. A count of `180000` hex = 1,572,864.
r = 1,572,864 / 16,777,216 = 0.09375; L = ln(0.90625 / 0.09375) = 2.26868
c0 + c1·L + c2·L² + c3·L³ = 0.0038077 → 1 / 0.0038077 = 269.55 K → **−3.600 °C**.

A count of 0 or 16,777,215 (FFFFFF) is an open or shorted thermistor and is removed as `sensor_error`. A channel whose unit is not *Degrees_C* is not a thermistor and is removed (`not_a_sensor`). Time stamps come from the header's logging start and sample period, re-anchored at every time-stamp event the logger wrote.

With `check_exports = TRUE` the `.hex` is compared with the Ruskin export of the same download at every shared time stamp: *agree* when the largest difference is under 0.0001 °C. In the example run all three RBR loggers agree to 5×10^{-8} °C.

The other loggers

Logger	What the app does	Removed as a sensor error
GeoPrecision FG2 .csv	Values are already °C. Columns `#1…#n` become `T1…Tn`; `HK-BAT` is battery (V). Tags such as `<NEW PAR>`, `<FIRMWARE RESET>`, `<MEMORY CLEAR>`, `<READOUT>` become events.	`(NoReply)`, `(Error)`: "logger reported (NoReply)"
HOBO U12 .csv	Values are already °C (HOBOware applies the TMC probe curve). The zone in the column header (`GMT-07:00`) is used.	blank cells (`missing_value`)
EM50 .xls	Processed sheet read; its *Unprocessed Data* sheet checked for error counts. Temperature, VWC and EC become separate variables per port (`P1_temp`, `P1_vwc`, `P1_ec`).	raw count ≥ 4094 ("raw count 4095 (sensor error code)"); VWC above 1
Campbell TOA5	Columns matched by the Campbell column patterns (e.g. `^T10[79]_C(_[0-9]+)?_Avg$` → temp; VWC, EC similarly).	`NAN`, `INF`, −7999, 7999, −6999, 6999, −9999

Example (EM50). Port 4 processed VWC 1.011 m³/m³ with raw count 4095 on the Unprocessed sheet → value removed, reason "raw count 4095 (sensor error code)". A processed VWC of 1.08 with no raw sheet → removed as "VWC above 1 (sensor error)".

---

3. Time stamps and time zones

All times are stored at **UTC−7** (`base_offset_h = -7`), with no daylight saving.

```
t_out = t_file − (offset_file − offset_base) hours
```

Which zone a file is in, first match wins:

1. A row of the time-zone table (`config/timezones.csv` : serial, model, file pattern, start, end, offset_hours)
2. A zone the file states: Ruskin `.txt` is UTC (0); a HOBOware header such as `GMT-07:00`
3. Otherwise the logger's own clock, taken as UTC−7 (GeoPrecision, EM50, RBR `.hex` / `.xls`, Campbell)

Examples. A Ruskin text row `2024-03-01T06:30:00` (UTC): 06:30 − (0 − (−7)) h = **2024-02-29 23:30**. A GeoPrecision row `15.08.2022 12:00:00` : no zone stated, no table row → **12:00**, unchanged. A HOBO file stating `GMT-06:00` : 12:00 → **11:00**.

> A logger set from a computer on daylight time is one hour off all year. Fix it with a time-zone table row for that serial (and dates), not by editing the file.

Standardising to UTC−6, and files that state no zone (added October 2026)

Every record is stored in the base zone, UTC−7 by default (Settings, *Base time zone*). Set it to −6 to standardise all records to UTC−6; nothing else changes by default. A file that states its zone is converted from that zone. A file that states none is read in the zone set under *Zone of files that state none*; left blank, it is read in the base zone, as before. If no file in a run states its zone and none has been chosen, the app asks which zone the loggers' clocks were in before running (a script carries on in the base zone and lists it in the problems). The time-zone table and a zone set by hand still win.

---

4. Clock faults and drift

Within each file the rows are checked in the order they were written:

Rule	When	Action
impossible_time	Before `min_year = 2000`, or more than 2 days in the future	removed
stray_record	A corrupt time stamp among good readings: off the time grid and out of the order the file was written in (`drop_stray_records = TRUE`, core `timestamps.R`)	removed; a reading written in order but off the grid is kept
clock_reset	The clock jumps back by more than a day compared with the latest trusted time so far (a firmware reset or flat clock battery)	removed: these readings cannot be placed in time
clock_resync	The clock was set back by less than a day; readings written earlier but stamped at or after a later reading's time	removed: the later readings are kept

```
reset    if  t_i < max(t_1 … t_(i−1), trusted) − 86400 s
resync   if  t_i ≥ min(t_(i+1) … t_n, trusted)
```

Examples from the example run. A GeoPrecision logger that lost its clock backup restarted at `1998-01-01 00:00:30` (the FG2 restart time): 1 row removed as *impossible_time*. An EM50 whose clock fell back to `2000-01-02 00:30` mid-record: 1,937 rows removed as *clock_reset*. GeoPrecision loggers set back at a readout: 10 readings per channel superseded (*clock_resync*).

RBR clock drift

The `.hex` header gives the computer's time and the logger's time at download. Whole hours (a zone difference) are removed, leaving the drift:

```
drift = (host − logger) − round((host − logger) / 3600) × 3600   s
with correct_rbr_drift = TRUE:
t' = t + drift × (t − t_start) / (t_logger_at_download − t_start)
```

Example. Host `15:47:04`, logger `15:32:13` → 891 s; round(891/3600) = 0 → drift = **+891 s** (14 min 51 s slow). With correction on, a reading halfway through the record moves +445 s, the last one +891 s. The default is `correct_rbr_drift = FALSE`: the drift is reported in `audit/clock.csv` but times are not changed.

Clock corrections by hand (`config/time_corrections.csv`: serial, start, end, shift_minutes, reason) shift every time of that serial in the window: `t' = t + shift_minutes × 60`, logged as *clock_shift_by_hand*.

5. Stitching each logger

All downloads of one logger are joined per channel and sorted by time. Values the logger marked as errors and blanks leave first. Where the same time appears in more than one download:

```
|v_new − v_old| < 0.000001  → duplicate_value: one copy kept
otherwise                  → overlap_conflict: the newest download is kept
```

Example. Download A (2023) and B (2024) both hold 2023-09-20 12:00 at 3.412 °C → one kept. If B had 3.415 °C, B's value is kept and A's is logged as *overlap_conflict* with the old value.

Gaps: a hole longer than `gap_factor = 3` × the logging interval (most common step) is listed in `audit/gaps.csv`. Nothing is interpolated.

Example. 30-minute logging; a gap is more than 90 minutes between readings. 12:00 then 14:00 → a 2.0 h gap.

6. Sensors, sites and depths

Every sensor (channel) gets a row in `config/sensors.csv` with site, depth and deployment start/end. Where the depth comes from, first match wins:

1. Set by hand, or seeded from the registry workbook (`depth_source` = what you typed, or *registry*)
2. A depth template for the make and model (`config/sensor_templates.csv`), e.g. Onset U12-008: 0.05, 0.20, 0.40, 1.00 m (*template*)
3. The default steps (*default*):

```
depth = default_first_depth + default_step × (n − 1)       0.10 m, 0.10 m
n = the number in the sensor name (T3 → 3, port P4 → 4; no number → 1)
```

Example. An RBR string with channels T1…T8 and no depths: T1 → 0.10 m, T5 → 0.50 m, T8 → **0.80 m**, all marked *default*. Because n comes from the name, a dead T3 that wrote nothing does not shift T4 to 0.30 m.

Assumed depths (*template*, *default*) are marked in every table and plot and are recomputed each run, so changing the steps applies at once. A logger not in the registry gets a site named after itself (e.g. *RBR_14040*).

Deployment window. A value outside its sensor's start and end is removed (`outside_deployment`). An end date without a time covers the whole day (+86,399 s).

Example. Start 2019-08-07; the logger was launched on 2019-07-08 at 12:52. The 708 hourly readings per channel before 7 August are removed, so a logger reading in the truck never reaches the site.

7. Quality rules and flags

Removed values are blanked and logged with the old value (they stay visible in the QC view). **Flagged** values are kept and marked in the `T_<depth>_flag` column and `audit/flags.csv`. Default rules (`config/qc_rules.csv` , page Quality rules):

Variable	Min	Max	Spike	Flat hours	Exempt band	Action	Why
temp	−50	50	1 °C			remove	open or shorted thermistor, failing logger
temp				720 (30 d)	0.2 °C	flag	stuck sensor; the zero curtain is exempt
vwc	0	0.8	0.3			remove	m^3/m^3
ec	0	20				remove	bulk EC, mS/cm (= dS/m)
battery	2.5	4.2				flag	GeoPrecision battery, V

A rule's `applies_to` can be *all*, a logger type, a serial or a site (several separated by |).

Limits

Example. An RBR node reads, say, −62.4 °C for one sample in winter: below −50 → removed as *range_min*. (In the example run, a few RBR nodes did this at a shallow plateau site.)

Spikes

A single reading that jumps away and comes straight back, by more than the rule's limit *and* far more than this sensor usually moves:

```
d1 = v_i − v_(i−1)        d2 = v_i − v_(i+1)
threshold = max( max_spike, spike_k × 1.4826 × MAD(d1) )      spike_k = 10
spike if sign(d1) = sign(d2) and min(|d1|, |d2|) > threshold
```

Example. A 0.55 m node whose steps have MAD 0.034 °C: threshold = max(1, 10 × 1.4826 × 0.034) = max(1, 0.51) = **1 °C**. Readings 5.20, 7.40, 5.25: d1 = +2.20, d2 = +2.15, same sign, both over 1 → 7.40 removed as *spike*. Readings 5.20, 7.40, 7.45 (a real step) → not a spike: d2 is −0.05.

Stuck sensor and the zero curtain

```
flat_line if a run of identical values lasts ≥ max_flat_hours  and  |value| >
flat_exempt_band
```

Example. A node reads exactly 3.412 °C for 31 days → flagged *flat_line*. A node reads −0.05 °C for 40 days in spring → exempt (|−0.05| ≤ 0.2): in freezing ground latent heat holds temperature at 0 °C for weeks.

Built-in checks (flags only)

Flag	When	Settings

air_like	At depth ≥ 0.2 m, a day's range (max − min of the raw readings, at least 4 readings) is over 15 °C: the sensor may be exposed, or above the ground	`check_air_like = TRUE`, `air_like_daily_range = 15`, `air_like_min_depth = 0.2`
pre_install	At the start of the record (within 60 days), every time stamp with ≥ 3 sensors has them all within 0.3 °C of each other, for at least 6 h: the string was not in the ground yet	`check_pre_install = TRUE`, `pre_install_spread = 0.3`, `pre_install_days = 60`
differs_from_other_sites	See section 13	`between_site_check = TRUE`
manual_flag / manual	Your entries in `manual_removals.csv` (site or serial, sensor, depth, variable, start, end, value range, action, reason, author)	Quality rules page

Example (air-like). A node at 1.00 m reads 3.1 °C at 05:00 and 19.6 °C at 15:00: range 16.5 > 15 → the whole day is flagged. **Example (pre-install).** Four nodes read 14.21, 14.30, 14.18, 14.25 °C (spread 0.12) for the first 45 hours after launch, then spread → those 45 hours flagged (as in the traced borehole).

Sensor health

```
failed        nothing usable kept
failing       errors > 50% of values read
intermittent  errors or QC removals > 2%
suspect       flagged > 10% of values kept
ok            otherwise
```

Example. The traced node: 90,445 read, 89,724 kept, 0 errors, 721 removed (0.8%), 45 flagged (0.05%) → **ok**, 99.2% valid.

8. Site records

A site holds **one instrument type**. A site name used for two types is split (e.g. *Plateau [RBR]* and *Plateau [HOBO]*), never merged. Loggers of the same type that replaced each other are joined; where two cover the same depth and time, the **later deployment** wins (`site_overlap`).

Example. Logger A (deployed 2019-08-07) and its replacement B (deployed 2024-09-21) both read 0.55 m on 2024-09-21 14:00 → B's value is kept and A's is logged.

Temperature goes to `sites/<site>.csv` (one column per depth, plus a flag column); moisture and conductivity go to `soil/<site>.csv` .

9. Daily means

```
expected  = round(86400 / logging interval s)
complete  if  n ≥ min_complete × expected        min_complete = 0.8
temp = mean, tmin = min, tmax = max of that day's readings (UTC−7 day)
```

Example. 30-minute logging → 48 per day, a day needs 38.4, so **39** readings. Hourly → 24, needs 20. A day with 36 of 48 readings is kept in `daily_means.csv` but `complete = FALSE` and is left out of every summary below.

10. Annual statistics and trumpet curves

From complete days only, per site, depth and year, both calendar year and water year (October to September, `water_year_start = 10`, named by the year it ends):

```
mean (MAGT), min, max of the daily means;  range = max − min
complete year if days ≥ year_complete × days in year      year_complete = 0.9
water year = calendar year + 1  if month ≥ October
```

Example. 0.55 m in 2022: 365 complete days ≥ 0.9 × 365 = 328.5 → complete; mean **3.139 °C**, min −10.471, max 16.868, range **27.338 °C**. 2019 (140 days, logger installed in August) → incomplete. 15 Nov 2022 → water year **2023**.

> For water years the leap-day test uses the water-year number (the year it ends), so a water year ending in a leap year is counted as 366 days.

The **trumpet curve** is these minima, means and maxima plotted against depth. The **depth of zero annual amplitude** is the shallowest depth whose annual range is under 0.1 °C in a complete calendar year (`zero_annual_amplitude.csv`; blank if no sensor is that deep).

Example. Ranges 30.6, 27.3, 18.6, 10.1 °C at 0.05–1.00 m: none under 0.1 → `zaa_m` blank. A 15 m borehole with a range of 0.08 °C at 12 m and 0.15 °C at 10 m → **12 m**.

11. Thaw depth and active-layer thickness

Each day, from complete daily means at depths ≥ 0, going down from the shallowest sensor, the first depth at or below 0 °C; the 0 °C crossing is interpolated linearly between it and the sensor above:

```
k = first sensor (going down) with T_k ≤ 0
thaw = z_(k−1) + (z_k − z_(k−1)) × T_(k−1) / (T_(k−1) − T_k)
```

Example (real, 10 May 2023). 0.05 m 11.598, 0.55 m 5.736, 0.85 m −0.029, 1.00 m 0.786 °C. First ≤ 0 is 0.85 m. thaw = 0.55 + 0.30 × 5.736 / (5.736 + 0.029) = 0.55 + 0.30 × 0.9949 = **0.848 m**.

Blank, with the reason in `state`, when the shallowest sensor is at or below 0 (*frozen at the shallowest sensor*), every sensor is above 0 (*thawed below the deepest sensor*), or there are fewer than two sensors.

> The search stops at the first frozen sensor. A frozen layer over warmer ground (as in this example, 1.00 m at +0.79 °C) is not seen as a talik; the thaw depth is the top of the frozen layer.

Active-layer thickness (ALT) per site and calendar year is the deepest daily thaw depth of the year, reported only where permafrost is present within the sensors:

```
permafrost  if some depth has annual max ≤ 0 °C (complete year)
ALT = max(thaw_m) over the year;  date_alt = its date
```

Example. A plateau borehole with nodes at 0.5, 1.0, 1.5, 2.0 m: annual max at 1.5 m is −0.3 °C → permafrost. On 2 Sept the 1.0 m node is +0.6 °C and 1.5 m −0.3 °C: thaw = 1.0 + 0.5 × 0.6 / 0.9 = 1.333 m, the deepest of the year → **ALT = 1.333 m**. In the example run no site had a depth frozen all year, so every result reads *no permafrost within the sensors* or *year incomplete*.

result	Meaning
ALT	Permafrost present, alt_m given
no permafrost within the sensors: thawed below the deepest sensor	The whole string thawed at some time
no permafrost within the sensors (no depth stays at or below 0 °C all year)	Seasonal frost only
year incomplete at one or more depths	Fewer than 90% of days at some depth
no thaw found	Permafrost, but no thaw depth between sensors (e.g. the top sensor never thawed)

12. Freeze/thaw dates and zero curtain

Per site, depth and water year, from complete daily means (`zero_band = 0.2` °C):

```
first_day_at_or_below_0   first day with T ≤ 0
last_day_at_or_below_0    last day with T ≤ 0
first_day_above_0_after_freezing   first day > 0 after that
days_frozen = days T < −0.2;  days_thawed = days T > 0.2;  days_zero_curtain = days |T| ≤ 0.2
zero-curtain period: ≥ zero_curtain_days (3) consecutive days with |T| ≤ 0.2
```

Example (real, 0.85 m, water year 2023). First ≤ 0 on 2022-11-05, last on 2023-05-23, thaw on 2023-05-24; 146 frozen, 156 thawed and 63 zero-curtain days. The periods 1–3 Apr (3 days, mean −0.18), 7–23 Apr (17 days) and 25 Apr–3 Jun (40 days, mean −0.005 °C) are listed in `zero_curtain.csv`. A missing day breaks a period.

13. Between-site check

For each depth that three or more sites share, each site's complete daily mean is compared with the median of the other sites that day. Sites differ by a steady, seasonal amount (snow, shade, landcover), so only a sudden departure from the usual offset counts:

```
offset    = T_site − median(T_others)
expected  = running median of offset over 31 days        between_site_window
departure = offset − expected
allow     = |others(t+3 d) − others(t−3 d)|               (fast spring/autumn
change)
flag if |departure| > max(6 × MAD(departure), floor) + allow
between_site_k = 6
floor = 5 °C at ≤ 0.1 m, 3 °C at ≤ 0.5 m, 1.5 °C deeper
```

Example (real). 0.55 m, 15 Aug 2022: site 16.750, others' median 15.910 → offset +0.840; expected +0.622 → departure **+0.219 °C**. Threshold ≥ max(6 × 0.42, 1.5) = 2.54 °C → not flagged. On 2–4 Dec 2020 the same site's 0.05 m node departed by +5.44 °C (a local snow or sensor difference: check the plot) → flagged for 3 days.

A summary per site and depth in `audit/between_sites.csv`: days compared, median offset, `r_change` (correlation of 7-day changes with the others'; not judged where the others change by under 0.2 °C), days departing and a verdict: *ok*, *departs at times*, *out of step* (r_change < `between_site_r_min = 0.5`), or *too short to judge* (< 30 days). Days are flagged, never removed: it may be a real local event.

14. Soil moisture, EC and battery

Volumetric water content (VWC, m^3/m^3) and bulk electrical conductivity (EC, mS/cm = dS/m) from EM50 ports or Campbell columns are checked with their own rules (VWC 0–0.8 with a 0.3 spike; EC 0–20) and written to `soil/<site>.csv` (columns `vwc_<depth>`, `ec_<depth>`). The app does not recalibrate them: values are as ECH2O Utility or the datalogger computed them.

```
soil_annual.csv: mean, min, max, n per site, variable, depth and calendar year
(all kept readings)
```

Example. A bog port at 0.1 m in 2021: mean VWC 0.132, min 0.062, max 0.800, 6,336 readings. A max of exactly 0.80 sitting at the rule limit suggests saturated or flooded readings were clipped: check the plot. A reading of 0.92 → removed as *range_max*.

Battery (GeoPrecision `HK-BAT`, V) is kept out of the site records and summarised per logger and day (`audit/battery.csv`: volts_min, volts_max); readings outside 2.5–4.2 V are flagged.

Example. 3.36 V at launch is a healthy lithium cell; a slow fall to 3.0 V over five years is normal; a drop under 2.5 V can explain a later clock reset to 1998.

15. Audit trail and reconciliation

```
values_read = values_out + removed      per logger and channel
(reconciliation.csv)
```

Every removal goes through one function that writes the stage, rule, site, logger, channel, variable, count and time span to `audit/audit_index.csv`, and each value (old value kept) to `audit/detail/` up to 200 values, else a summary line. The run stops if any row does not reconcile.

> **Example.** The traced logger's file: 376,864 values read (4 channels + battery) = 373,949 in the output (site + battery table) + 2,915 removed (2,832 outside deployment, 42 clock resync, 41 spikes) → reconciles.

16. WMO quality codes in exports

Exports give each value a `qc` column (long layout) or a `T_<depth>m_qc` column (wide layout) with the code of WMO BUFR table 0 33 020, "Quality control indication of following value" (WMO Codes Registry): 0 good, 1 inconsistent, 2 doubtful, 3 wrong, 4 not checked, 5 has been changed, 6 estimated, 7 missing value. The `flag` text still names the checks. A string that was not yet in the ground (`pre_install`) is 3; an air-like swing, a flat run, a spike or a difference from the other sites is 2; temperature with no quality rule is 4. When several checks apply the worst is given, and an hourly, daily or monthly mean takes the worst code of the values in it. *Checks only* on the export page leaves the `qc` column out.

Reading the outputs

File	What it holds
sites/<site>.csv	Temperature, one column per depth (`T_0.55m`) and its flag column, times at UTC−7
soil/<site>.csv	VWC and EC per depth
summaries/daily_means.csv	site, depth_m, date, temp, tmin, tmax, n, expected, complete
summaries/annual_stats.csv	Calendar and water year: mean, min, max, range, days, dates of min and max, complete
summaries/zero_annual_amplitude.csv	zaa_m per site and year
summaries/thaw_depth_daily.csv	thaw_m and state per site and day
summaries/active_layer.csv	alt_m, date_alt, permafrost, deepest frozen depth, result
summaries/freeze_thaw.csv, zero_curtain.csv	Freeze/thaw dates and day counts per depth and water year; zero-curtain periods
summaries/soil_annual.csv	Soil moisture and EC per year
audit/files.csv, bundles.csv, file_audit.csv	Every file: format, status, zone and why, values read, kept, dropped and why
audit/clock.csv, events.csv	Clock resets, resyncs, impossible times, RBR drift, hand shifts; logger tags
audit/export_checks.csv	RBR .hex against its Ruskin export
audit/flags.csv	Every flagged period: site, logger, channel, start, end, records, check, detail, then `rule_id` (the row of qc_rules.csv, or the check's name), `qc_code` (WMO), `app`, `run_id`
audit/sensor_health.csv, gaps.csv	Per sensor: read, kept, errors, removed, gaps, % valid, health

audit/between_sites.csv	Between-site summary per site and depth
audit/audit_index.csv, reconciliation.csv	Every removal; read = out + removed per channel
audit/sites_summary.csv, loggers.csv, battery.csv, problems.csv	Catalogue of sites and loggers, battery per day, everything that needs attention
report.html	The run report; site reports from Reports and export

Sensor notes

From the makers' manuals and data sheets. The accuracy of a ground temperature record is usually limited by the thermistor and its depth, not the logger.

Logger / sensor	Range	Accuracy	Resolution	Clock and memory
RBR XR-420 thermistor string (TR-1050 / T8 type)	−5 to 35 °C standard (wider on request)	±0.002 °C (factory calibrated)	<0.00005 °C	about ±1 min/yr; drift recorded at download
RBR concerto³ / duo³ (successor; Ruskin)	−5 to 35 °C standard (wider on request)	±0.002 °C initial; stability ±0.002 °C/yr	<0.00005 °C	±60 s/yr; 240 M readings; 8 AA cells
GeoPrecision FG2 / M-Log5W with TNode thermistor chain	about −40 to 60 °C	typically ±0.1 °C (TNI sensors; ±0.05 °C with calibration)	0.01 °C	radio readout (cryoDongle / FlexGate, 10–100 m); append-only flash; after losing clock backup restarts at 1998-01-01 00:00:00; logs battery and internal temperature
Onset HOBO U12-008 (4-channel external)	0–2.5 V inputs, 12-bit; logger −20 to 70 °C	±2 mV ±2.5% of reading (voltage)	0.6 mV	±1 min/month at 25 °C; 43,000 measurements; 1-year battery
Onset TMCx-HD thermistor probe (with U12)	−40 to 50 °C in water/soil	±0.25 °C from 0 to 50 °C (larger below 0)	0.03 °C at 20 °C	response 30 s in stirred water
METER/Decagon EM50 with 5TM / 5TE / ECT	logger −40 to 60 °C; 5 ports	VWC ±0.03 m³/m³ generic (±0.01–0.02 soil-specific); temperature ±1 °C; 5TE EC ±10% (0–7 dS/m)	VWC 0.0008 m³/m³, 0.1 °C, EC 0.01 dS/m	36,864 scans; overwrites oldest when full; 5 AA; raw 4094/4095 = error
Campbell 107 / 109 probes (TOA5 tables)	107: −35 to 50 °C; 109: −50 to 70 °C	±0.2 °C typical interchangeability (worst case up to ±0.5 °C at the range ends)	set by the datalogger (<0.01 °C)	datalogger clock; NAN and ±7999 are out-of-range codes

Known quirks

- **GeoPrecision clock resets.** A flat or replaced battery restarts the clock at 1998-01-01; those rows are removed as *impossible_time* or *clock_reset*. Readings after a reset up to the next readout cannot be placed in time unless you know the offset (then use a clock correction).

- **GeoPrecision resync at readout.** When the dongle sets the clock back a few minutes, a few readings overlap in time; the later ones are kept (*clock_resync*).
- **.f2b and .hobo cannot be read.** Always export them (FG2-Shell CSV, HOBOware CSV). A bundle with no export is listed as *needs export*.
- **Ruskin text is UTC**; the .hex is logger time. The app handles both; never mix a hand-shifted export with the raw file.
- **RBR drift** is reported, not corrected, by default. Over a 3-year deployment, 15 minutes of drift matters little for daily means.
- **HOBO TMC probes** are less accurate below 0 °C than above it; the zero curtain may read a few tenths off zero.
- **EM50 memory** wraps: the oldest scans are overwritten when full, so download in time: 36,864 scans last about 768 days at 30-minute logging, half that at 15 minutes. Its clock falls back to 2000-01-01 after a battery change.
- **Self-heating and lag.** Ground sensors lag the air by hours to months with depth. A shallow sensor in a dry, open hole can read like air: see *air_like*.

Reasonable values

Typical ranges for subarctic and arctic boreholes in peatland and tundra with discontinuous permafrost. Use them to spot problems, not as limits.

Parameter	Typical	Possible but check	Not possible / sensor fault
Near-surface (≤ 0.1 m) ground temperature	−25 to 25 °C	−35 to −25, 25 to 35 °C (thin snow, bare dark ground)	< −50 or > 50 °C
Ground at 0.5–1 m	−15 to 18 °C	−20 to −15 °C	a daily range > 15 °C (air-like)
Ground at 5–15 m	−5 to 5 °C, nearly constant	annual range 0.1–1 °C	daily swings of degrees
Daily range at 0.05 m	< 10 °C	10–20 °C (bare, sunny)	> 25 °C under snow
Step between readings (30 min, ≥ 0.5 m)	< 0.1 °C	0.1–1 °C (rain, meltwater)	a jump > 1 °C that comes straight back (spike)
MAGT, permafrost	−3 to 0 °C	−6 to −3 °C (colder north)	
Zero curtain	days to 8 weeks, |T| ≤ 0.2 °C	> 8 weeks in wet peat	exactly constant to 0.001 °C for months (stuck)
Active-layer thickness	0.4–1.2 m (peat plateaus, tundra)	1.2–3 m (mineral soil, burned)	deeper than the sensors (reported as such)
Thaw depth day to day	< 2 cm/day	2–5 cm/day in early summer	jumps back and forth by tens of cm
VWC (peat, unfrozen)	0.3–0.8 m^3/m^3	0.05–0.3 (dry surface, frozen: liquid water only)	< 0 or > 0.8; raw 4095
Bulk EC (peat)	0.01–0.5 mS/cm	0.5–2 mS/cm	< 0 or > 20
GeoPrecision battery	3.0–3.6 V	2.5–3.0 V (replace soon)	< 2.5 V
Between-site departure	< 1 °C	1–3 °C at the surface	persistent > 3 °C at depth

Using the app: the set-up flow and the Workbench

The app has two views of the same project, data and edits: the **Standard** view, a numbered set-up flow (1 Bring in, 2 Organize, 3 Correct, 4 Report, 5 Save), and the **Workbench**, a chart over a grid of every value. The switch at the top right moves between them; an edit made in either shows in the other at once.

Check the data

- Each site or station gets a plot of its main series with what the automatic checks found listed under it. Decide each one (Keep, Remove, or Flag where offered); the decisions wait in the "unsaved changes" chip at the top until **Apply**, then one run applies them.
- Drag across a plot with the right mouse button (or shift-drag) to select a stretch: its records, missing values, mean, minimum, maximum and change are listed under the plot. **Look closer** zooms to it and opens the Workbench there.
- The view bar sets every plot to half-hourly, as recorded, hourly, daily, monthly or annual values; the smoothing and the ghost of the raw data are display only.

The Workbench

- The tree on the left picks a site, station, deployment or table and its parameters; the chart shows one panel per parameter, the grid every value with its unit under the heading.
- Dashed red lines on a panel are the quality-rule limits (Settings: quality rules) when they fall within the view; each panel is labelled with its unit.
- Right-drag on the chart (or drag in the grid) selects values; each further right-drag adds to the selection, Esc clears it. The inspector on the right lists the selection's count, mean, standard deviation, minimum, maximum, missing, removed and flagged values.
- **Remove**, **Flag** and **Restore** write rows to the set-up tables (manual removals) with a reason and initials. They show as pending until the next run applies them; the old values stay in the audit trail and raw files are never changed.

When something goes wrong

- If part of a page fails, a notice says so with a short reference (E- and six characters) and the app carries on. The life-ring button collects versions, logs and the set-up into a troubleshooting zip (no raw data) to send with the reference.
- If the output folder is deleted or moved while the app is open, the pages notice within seconds and show what is left; run again to rebuild it.
- A project folder that cannot be written to (read-only, a locked or full drive), or a file where the output, cache or config folder should be, stops the run before anything is changed, with what to do.
- A set-up table that is damaged or was saved by another program with columns missing is shown with those columns added empty and a notice; nothing changes in the file until it is saved.

Troubleshooting

Message or symptom	What it means	What to do
needs_export: Open the .f2b in FG2-Shell ...	A download with no readable file	Export the CSV into the same folder and run again

unsupported: mixed-array / binary	A Campbell table GT cannot read	Convert to TOA5 with CardConvert
Many *outside_deployment* removals	Deployment start/end in sensors.csv too narrow, or the wrong year	Check dates on Sensors and depths
Site named after a logger (*RBR_14040*)	Serial not in the registry	Give it a site on Sensors and depths
Depths marked default or template	No measured depth for the sensor	Enter measured depths; summaries depend on them
clock_reset with hundreds of rows	The logger lost its clock	If the true offset is known, add a clock correction; otherwise those rows stay out
A whole-hour step at a download	A file in a different zone	Add a row to the time-zone table
export_checks: disagree or no matching times	.hex and Ruskin export differ, or the export is in another zone	Re-export; check the zone
pre_install flag at the start	String logging before it was in the ground	Trim on Plot and explore, or move the deployment start
air_like at depth	Sensor exposed, frost-heaved or at the wrong depth	Check field notes; remove the period by hand if exposed
flat_line flag	Stuck sensor or logger repeating its last value	Check in the QC view; remove by hand if confirmed
ALT blank: no permafrost within the sensors	No depth stays ≤ 0 °C all year	Correct for seasonal frost; deeper sensors needed for ALT
Thaw depth stuck just above a sensor	That sensor is in a long zero curtain	Expected; the 0 °C crossing interpolates to that node
between sites: out of step	A site does not follow the others	Check depth, clock and site assignment
Run stopped: values do not reconcile	A count does not add up (a bug or a file changed mid-run)	Run again; if it persists, send the audit folder

Glossary

Active layer, ALT	Ground above permafrost that thaws each summer; its thickness is the year's deepest thaw.
Bundle	The files of one download: the logger's own file and its exports.
Clock reset / resync	The logger clock jumping back by more than a day / set back by less.
Complete day / year	A day with ≥ 80% of its readings; a year with ≥ 90% of its days.
Flag	A mark on a kept but doubtful value.
MAD	Median absolute deviation; × 1.4826 approximates a standard deviation, robust to outliers.
MAGT	Mean annual ground temperature at a depth.
Permafrost	Ground at or below 0 °C for at least two consecutive years (here: a sensor with annual max ≤ 0 °C).
Talik	An unfrozen layer within or below frozen ground.
Thermistor string / chain	Several temperature sensors on one cable at set depths.

Trumpet curve	Annual minimum, mean and maximum plotted against depth.
UTC−7	The single time zone all data are stored in (no daylight saving).
VWC, EC	Volumetric water content (m^3/m^3); bulk electrical conductivity (mS/cm).
Water year	October to September, named by the year it ends.
Zero curtain	A period when ground holds near 0 °C while water freezes or thaws (latent heat).
ZAA	Depth of zero annual amplitude: where the annual range falls below 0.1 °C.

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